

Homework cover page

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Problem set no: 7

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Homework 7

1)

know: $q''_s = 10,000 \text{ W/m}^2$, $\rho = 600 \text{ kg/m}^3$, $k = 1.5 \text{ W/mK}$, $c_p = 3000 \text{ J/kgK}$

$M = 1.5 \times 10^3 \text{ Pa}$, $b = 10 \text{ cm}$, $\dot{m} = 0.25 \text{ kg/s}$, $T_m = 400^\circ\text{C}$, $T_{h,i} = 450^\circ\text{C}$

find: a) L , q

b) h

c) T_{sur} @ $x=0$, $x=L$, sketch of temp profile.

Assumptions:

Analysis: uniform fluid properties, therm. fully developed

a) $q = \dot{m} c_p \Delta T = 0.25 \text{ kg/s} (3000 \text{ J/kgK}) (450^\circ\text{C} - 400^\circ\text{C}) = 37.5 \text{ kW}$

$37.5 \text{ kW} = q'' A$

$A = 2\pi r L = \pi b L$, $b = \frac{10}{100} \text{ m}$

$37.5 \cdot 10^3 \text{ W} = 10000 \text{ W/m}^2 (\pi \cdot \frac{10}{100} \text{ m} L)$

$L = 11.94 \text{ m}$

$\dot{m} = \rho v A_c$

b) $Re_b = \frac{\rho v b}{\mu} = \frac{\dot{m} b}{\mu A_c} = \frac{\dot{m} b}{\mu (\frac{\pi b^2}{4})} = \frac{4 \dot{m}}{\mu \pi b} = 202.066$

$h = \frac{Nu k_f}{L} = \left(\frac{0.664 Re_b^{1/2} Pr^{1/4}}{L} \right) \cdot k_f = Nu (q''_s = \text{const}) = \frac{4.36 \cdot k_{fp}}{b} = 65.9 \text{ W/m}^2\text{K}$

from Handwritten note 33.pdf

c) $q'' = h \Delta T$

$q'' = h(T_x - T_i)$

$\frac{q''}{h} + T_i = T_x$

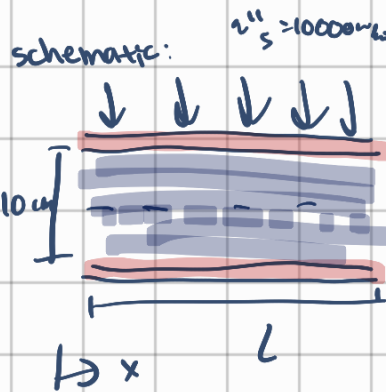
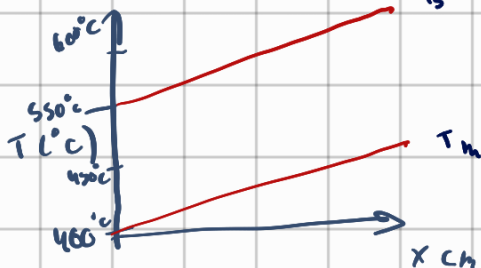
$T: (x=0)$

↓

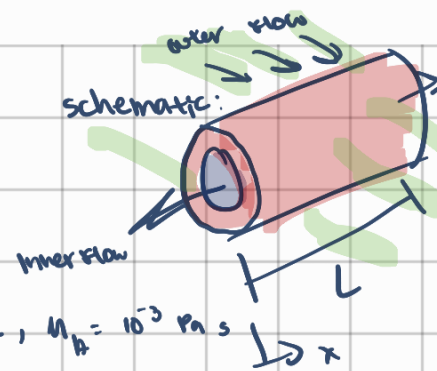
@ $x=0$: $\frac{10,000 \text{ W/m}^2}{65.9 \text{ W/m}^2\text{K}} + 400 = 552.9^\circ\text{C}$

$T_i(x=L) = 602.9^\circ\text{C}$

@ $x=L$: $\frac{10,000 \text{ W/m}^2}{65.9 \text{ W/m}^2\text{K}} + 450$



2)



know: $\dot{m} = 0.05 \text{ kg/s}$, $b = 0.1 \text{ m}$, $T_{m,i} = 85^\circ\text{C}$, $L = 5 \text{ m}$, $U_\infty = 5 \text{ m/s}$, $T_\infty = 10^\circ\text{C}$, $\mu_f = 10^{-3} \text{ Pa}\cdot\text{s}$

$k_A = 0.7 \text{ W/m}\cdot\text{K}$, $c_p = 500 \text{ J/kg}\cdot\text{K}$, $\nu_b = 20 \cdot 10^{-6} \text{ m}^2/\text{s}$, $k_b = 0.03 \text{ W/m}\cdot\text{K}$, $Pr = 0.7$

find: a) $Re_{b,i}$

b) Nu_b, h_i

c) $\bar{h}_o$

d) mean T @ outlet, temp profile

e) $\dot{q}_{tot}$

Assumptions: uniform material properties, thermally developed

Analysis: 9.2

a) $Re = \frac{\dot{m} b}{\mu_f} = \frac{0.05 \text{ kg/s} \cdot 0.1 \text{ m}}{10^{-3} \text{ Pa}\cdot\text{s} \cdot \frac{\pi}{4} b^2} = 635.6 : \text{laminar}$

b) $Nu_b = 4$

$h = \frac{Nu_b k}{b} = \frac{4 \cdot 0.7 \text{ W/m}\cdot\text{K}}{0.1 \text{ m}} = 28 \text{ W/m}^2\cdot\text{K}$

c) $Re_{outer} = \frac{U_\infty b}{\nu_b} = \frac{5 \text{ m/s} \cdot 0.1 \text{ m}}{20 \cdot 10^{-6} \text{ m}^2/\text{s}} = 25000 \text{ (turbulent)}$

$Nu_{cylinder} = C Re^m Pr^n$
 $Nu_{cylinder} = 0.183 (25000)^{0.6} (0.7)^{1/3} = 89.5$

$h = \frac{Nu_b k}{b} = \frac{89.5 \cdot 0.03 \text{ W/m}\cdot\text{K}}{0.1} = 26.85 \text{ W/m}^2\cdot\text{K}$

d) $\bar{R}_{cond} = \frac{1}{h_{cond}}$

$\frac{1}{\bar{h}_{tot}} = \frac{1}{h_A} + \frac{1}{h_b} \Rightarrow \bar{h}_{tot} = \left(\frac{1}{28} + \frac{1}{26.85} \frac{\text{m}^2\cdot\text{K}}{\text{W}} \right) = 13.706 \text{ W/m}^2\cdot\text{K}$

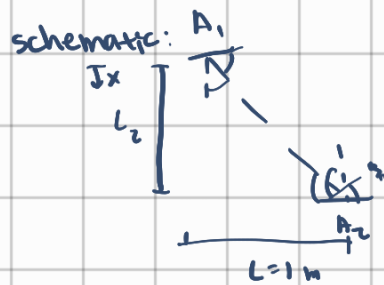
$\frac{T_\infty - T_{b,e}}{T_\infty - T_{m,i}} = e^{\frac{\bar{h} p L}{\dot{m} c_p}}$

$\frac{10^\circ\text{C} - T_{b,e}}{10^\circ\text{C} - 85^\circ\text{C}} = e^{\frac{13.7 \text{ W/m}^2\cdot\text{K} \cdot \pi \cdot 0.1 \text{ m} \cdot 5 \text{ m}}{0.05 \text{ kg/s} \cdot 500 \text{ J/kg}\cdot\text{K}}} = 91.7^\circ\text{C}$

e) $\dot{q} = \dot{m} c_p \Delta T = 0.05 \text{ m/s} (500 \text{ J/kg}\cdot\text{K}) (85^\circ - 91.7^\circ\text{C}) = 1082.5 \text{ W}$

3)

know: $A_1 = 10^{-4} \text{ m}^2$, $A_2 = 2 \times 10^{-4} \text{ m}^2$, $I_1 = 10000 \text{ W/m}^2 \cdot \text{sr}$



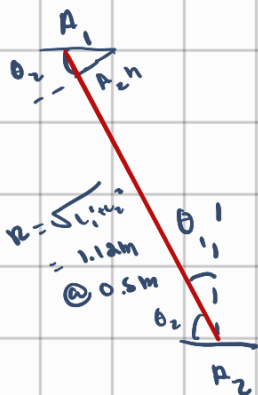
find: a) Irradiation of A_2 for $L_2 = 0.5 \text{ m}$

b) plot irradiation of A_2 over $0 < L_2 < 10 \text{ m}$

Assumptions: uniform mat. prop., diffuse emission

Analysis:

a) $I_2 = 10000 \text{ W/m}^2 \cdot \text{sr}$, $r = \sqrt{1^2 + 0.5^2} \text{ m} = 1.12 \text{ m}$



$$\theta_2 = \tan^{-1}\left(\frac{L_2}{L_1}\right) = 26.6^\circ$$

$$\theta_1 = 90 - 26.6^\circ = 63.4^\circ$$

$$dA_n = r^2 d\omega \rightarrow d\omega = \frac{dA_n}{r^2}$$

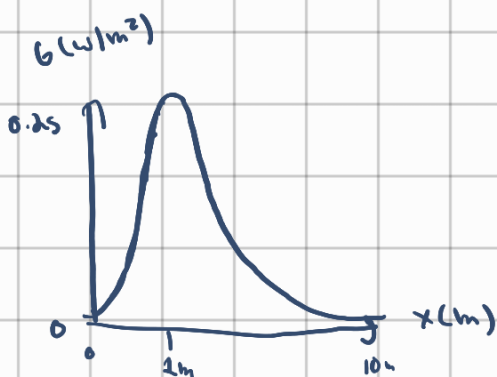
$$\cos^2 \theta = A \cos \theta$$

$$G = \int_0^\infty \pi I_{\lambda,i}(\lambda) d\lambda = \int \pi \left(\frac{d\omega}{dA_1 \cos \theta} \right) d\lambda \times$$

$$G = \int_0^\infty \int_0^\infty \int_0^\infty I_{\lambda,i}(\lambda, \theta, \phi) \cos \theta \sin \theta d\theta d\phi d\lambda$$

$$G = \frac{I_1 A_1 \cos \theta}{A_2} = \frac{I_1 A_1 \cos \theta \cdot A_2 \cos \theta}{A_2 r^2} = \frac{10000 \cdot 10^{-4} \cos(63.4^\circ)^2 \cdot 2 \cdot 10^{-4}}{2 \cdot 10^{-4} \cdot (1.12^2)} = 0.16 \text{ W/m}^2$$

$$b) A_1 = x: G = \frac{I \cdot A_1 \cdot \sin\left(\tan^{-1}\left(\frac{L_2}{L_1}\right)\right)^2}{A_2 x^2} = \frac{I A_1 x^2}{(1+x^2)^2}$$



1)

know: $T_s = 2000 \text{ K}$, $T_{\text{sur}} = 400 \text{ K}$, $\epsilon_{\lambda, \theta}$ vs λ plot

Table 12.2 from page 720

find: a) emissive power from $1 \mu\text{m}$ to $3 \mu\text{m}$ $0 \leq \theta \leq 60^\circ$

b) ϵ , α ^{tot, hemisphere}

c) E , G , J , ϵ''_{rad}

Assumptions: opaque, uniform material properties

Analysis:

a) $E = \epsilon \sigma T^4$

$$\lambda_1 T @ 1 \mu\text{m} = 2000 = 0.066728$$

$$\lambda_2 T @ 1.5 \mu\text{m} = 3000 = 0.273232$$

$$\lambda_3 T @ 3 \mu\text{m} = 6000 = 0.7378$$

$$\epsilon_m = 0.2 (0.273232 - 0.066728) + 0.8 (0.7378 - 0.273232) = 0.413$$

$$E = \epsilon_{\text{tot}} \sigma T_s^4, T_s = 2000 \text{ K}, \sigma = 5.67 \cdot 10^{-8} \text{ W/m}^2 \text{K}^4$$

$$E = 3.75 \text{ ES W/m}^2$$

for 60° :

$$3.75 \cdot 10^5 \cdot \sin(60)^\circ = 2.8 \text{ ES W/m}^2$$

b) for $T = 400$

$$\epsilon = \epsilon_{\lambda_1} F_{0 \rightarrow \lambda_1} + \epsilon_{\lambda_2} (F_{0 \rightarrow \lambda_2} - F_{0 \rightarrow \lambda_1}) + \epsilon_{\lambda_3} (1 - F_{0 \rightarrow \lambda_2})$$

$$\epsilon = 0.2 (0.273232) + 0.8 (1 - 0.273232) = 0.636$$

$$\alpha = \frac{0.25(0) + 0.8(2-0)}{1} = 0.8$$

$$c) E = \epsilon \sigma T_s^4 = 0.636 (5.67 \cdot 10^{-8}) (2000^4) = 5.97 \cdot 10^5 \text{ W/m}^2$$

$$G = \sigma T^4 = 5.67 \cdot 10^{-8} (400^4) = 1479.68 \text{ W/m}^2$$

$$J = \epsilon + (1 - \alpha) G = 5.97 \cdot 10^5 + 0.25 (1479.68) = 5.98 \cdot 10^5 \text{ W/m}^2$$

